

5. Ranking Sorting Algorithms

Fastest (1) to Slowest (6)

1. Merge Sort
2. Quick Sort
3. Shell Sort
4. Insertion Sort
5. Selection Sort
6. Bubble Sort

Compared to the other sorting algorithms, Merge Sort and Quick Sort have a best case time complexity of $O(N \log N)$, which is faster than the other algorithm's run time. The reason why I had Merge Sort above Quick Sort is because Merge Sort, in both the best and worst case, is $O(N \log N)$. On the other hand, Quick Sort's worst case, although it rarely occurs, is $O(N^2)$, which is slower than $O(N \log N)$. Also, both use recursion for sorting and split an array into two arrays to sort, making them more efficient than the rest.

The slowest sorting algorithm is Bubble Sort, because of its seemingly redundant approach to sorting. With a time complexity of $O(N^2)$, it sorts through an array one at a time, comparing the current to the next element, then repeating until all are sorted.

Selection and Insertion Sort are nearly the same in time complexity, but Inserting is easier to perform and quicker than Selecting an element from an array.

Shell Sort is ranked third because it's better than Insertion Sort, but not as efficient as Quick Sort.